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Course Name – Data Analytics With GenAI

Subject – Statistics

Topic – Descriptive Statistics

Q1: Understanding Central Tendency (Easy)

A bakery tracks the daily sales of muffins (in dozens) over a week: [10, 12, 11, 15, 14, 13, 12]. What is the most representative value of their weekly sales, and why?

Solution:

Given data: [10, 12, 11, 15, 14, 13, 12]

Step 1: Arrange the data in order from smallest to largest

- 10, 11, 12, 12, 13, 14, 15

Step 2: Find the median (middle value)

- Count total values: 7 values
- Middle position: $(7 + 1) \div 2 = 4\text{th position}$
- The 4th value is **12**

Answer: The most representative value is 12 dozens, which is the median. This is the best measure because it indicates the center of the data. It shows that on a typical day, the bakery sells about 12 dozens of muffins.

Q2: Mean in Real Life (Easy)

A teacher records the marks of her students in a short quiz: [12, 15, 14, 16, 18, 20, 19]. What is the mean score, and what does it tell us about the class's performance?

Solution:

Given data: [12, 15, 14, 16, 18, 20, 19]

Step 1: Add all the scores together

- $12 + 15 + 14 + 16 + 18 + 20 + 19 = 114$

Step 2: Count how many scores there are

- Total students: 7

Step 3: Divide the sum by the number of scores

- Mean = $114 \div 7 = 16.29$ (approximately **16.3**)

Answer: The average score is 16.3 marks. This means the class did fairly well on the quiz, with most students scoring above the middle range. The class has a good overall performance.

Q3: Mode in Real Life (Easy)

A store records the shoe sizes sold in one day: [7, 8, 9, 8, 8, 10, 7, 9].

What is the mode, and why is this information useful for the store manager?

Solution:

Given data: [7, 8, 9, 8, 8, 10, 7, 9]

Step 1: Count how many times each size appears

- Size 7: appears 2 times
- Size 8: appears 3 times
- Size 9: appears 2 times
- Size 10: appears 1 time

Step 2: Find which size appears most often

- Size **8** appears the most (3 times)

Answer: The mode is size 8. This helps the store manager know which shoe size to stock the most.

Because size 8 sells the most frequently, they should keep plenty of size 8 shoes in inventory to meet customer demand and avoid running out of this popular size.

Q4: Median in Real Life (Medium)

A car dealer notes the prices of used cars: [\$8,000, \$9,500, \$10,200, \$11,000, \$50,000]. Why is the median a better measure than the mean in this case? Calculate the median.

Solution:

Given data: [\$8,000, \$9,500, \$10,200, \$11,000, \$50,000]

Step 1: The data is already arranged in order

Step 2: Find the median (middle value)

- Count: 5 values
- Middle position: $(5 + 1) \div 2 = 3\text{rd position}$
- The 3rd value is **\$10,200**

Step 3: Calculate the mean for comparison

- Sum: $\$8,000 + \$9,500 + \$10,200 + \$11,000 + \$50,000 = \$88,700$
- Mean: $\$88,700 \div 5 = \mathbf{\$17,740}$

Answer: The median is \$10,200. The median is a better measure than the mean, which is \$17,740, because there is one very high value, \$50,000, that is an outlier. This outlier raises the mean, making it unrepresentative of most car prices. The median is unaffected by this extreme value, providing a clearer picture of the typical car price at this dealership.

Q5: Dispersion Introduction (Medium)

A student times how long it takes to finish a puzzle each day: [25, 30, 27, 35, 40]. What does the range tell us about the variation in the student's puzzle-solving time?

Solution:

Given data: [25, 30, 27, 35, 40]

Step 1: Find the smallest value

- Minimum = 25 minutes

Step 2: Find the largest value

- Maximum = 40 minutes

Step 3: Calculate the range

- Range = Maximum - Minimum
- Range = $40 - 25 = \mathbf{15 \text{ minutes}}$

Answer: The range is 15 minutes. This shows that there is a lot of variation in how long it takes the student to solve the puzzle each day. The student's time can change by as much as 15 minutes, which suggests that their puzzle-solving speed is not very consistent. It may depend on factors like the difficulty level, concentration, or fatigue.

Q6: Range in Action (Medium)

A farmer records the weekly weight of harvested apples (kg): [100, 105, 98, 110, 120]. Find the range. How can this help the farmer in planning his packaging?

Solution:

Given data: [100, 105, 98, 110, 120] kg

Step 1: Find the smallest weight

- Minimum = 98 kg

Step 2: Find the largest weight

- Maximum = 120 kg

Step 3: Calculate the range

- Range = Maximum - Minimum
- Range = $120 - 98 = 22 \text{ kg}$

Answer: The range is 22 kg. This helps the farmer with planning and packaging because it shows there is a lot of variation in apple weights from week to week. The farmer needs to prepare packaging that can hold apples weighing between 98 kg and 120 kg. They may also need to consider different storage space and transportation capacity to manage this 22 kg difference in weekly harvests.

Q7: Variance for Decision-Making (Medium)

Two delivery companies track delivery delays (in minutes). Company A: variance = 6

Company B: variance = 15

Which company is more consistent, and why?

Solution:

Given information:

- Company A: variance = 6
- Company B: variance = 15

Step 1: Understand what variance means

- Variance measures how spread out the data is from the average
- Lower variance = more consistent (less variation)
- Higher variance = less consistent (more variation)

Step 2: Compare the variances

- Company A has variance of 6 (lower)
- Company B has variance of 15 (higher)

Answer: Company A is more consistent. Company A has a lower variance of 6, while Company B's variance is 15. This means Company A's delivery delays vary less from their average. In practical terms, Company A's delivery times are more predictable and reliable. If you need to plan around delivery times, Company A would be the better choice because you can more accurately predict when your package will arrive. Company B's higher variance makes their delivery times more unpredictable, with greater differences from one delivery to another.

Q8: Standard Deviation in Context (Hard)

A finance student compares the daily price fluctuations of two cryptocurrencies.

Coin A: standard deviation = \$30 Coin B: standard deviation = \$120. Which coin is riskier to invest in, and why?

Solution:

Given information:

- Coin A: standard deviation = \$30
- Coin B: standard deviation = \$120

Step 1: Understand what standard deviation means

- Standard deviation measures how much prices vary from the average

- Higher standard deviation = more volatility/risk
- Lower standard deviation = more stability/less risk

Step 2: Compare the standard deviations

- Coin A: SD = \$30 (lower volatility)
- Coin B: SD = \$120 (higher volatility)

Step 3: Analyze the risk

- Coin B's standard deviation is 4 times larger than Coin A's (\$120 vs \$30)
- This means Coin B's price swings are much more dramatic

Answer: Coin B is a riskier investment. Its standard deviation is \$120, while Coin A's is only \$30. This means Coin B's daily price changes a lot more. A higher standard deviation shows greater volatility and unpredictability. While this unpredictability could lead to bigger gains, it also raises the chance of major losses. Coin B's price could move up or down by \$120 or more from its average, making it much riskier for anyone seeking stable, predictable returns.

Q9: Combining Measures (Hard)

A family records their monthly electricity usage (in kWh): [400, 420, 390, 450, 410].

Find the mean and standard deviation. What do these values together tell you about the family's energy use pattern?

Solution:

Given data: [400, 420, 390, 450, 410] kWh

Step 1: Calculate the Mean

Add all values:

- $400 + 420 + 390 + 450 + 410 = 2,070$ kWh

Divide by the number of months:

- Mean = $2,070 \div 5 = 414$ kWh

Step 2: Calculate the Standard Deviation

Find the deviation of each value from the mean:

- $(400 - 414) = -14$
- $(420 - 414) = 6$
- $(390 - 414) = -24$
- $(450 - 414) = 36$
- $(410 - 414) = -4$

Square each deviation:

- $(-14)^2 = 196$
- $(6)^2 = 36$
- $(-24)^2 = 576$
- $(36)^2 = 1,296$
- $(-4)^2 = 16$

Sum the squared deviations:

- $196 + 36 + 576 + 1,296 + 16 = 2,120$

Divide by the number of values:

- Variance = $2,120 \div 5 = 424$

Take the square root:

- Standard Deviation = $\sqrt{424} = \mathbf{20.6 \text{ kWh}}$ (approximately)

Answer:

- **Mean = 414 kWh**
- **Standard Deviation = 20.6 kWh**

What these values tell us: The family uses an average of 414 kWh per month, which gives them a baseline for budgeting their electricity costs. The standard deviation of 20.6 kWh shows there is moderate variation in their monthly usage. This relatively low standard deviation (about 5% of the mean) indicates the family has fairly consistent energy usage habits from month to month. However, there is one month (450 kWh) where usage was noticeably higher, which might be due to seasonal factors like air conditioning use or having guests. Overall, their energy consumption is predictable and stable.

Q10: Practical Application (Hard)

A basketball player's points in 8 games are recorded: [15, 18, 20, 22, 25, 17, 19, 21].

Find the mean, median, mode, range, and standard deviation. What insights can these measures provide about the player's scoring performance?

Solution:

Given data: [15, 18, 20, 22, 25, 17, 19, 21] points

Step 1: Calculate the Mean

Add all values:

- $15 + 18 + 20 + 22 + 25 + 17 + 19 + 21 = 157$ points

Divide by number of games:

- Mean = $157 \div 8 = \mathbf{19.625 \text{ points}}$ (approximately **19.6 points**)

Step 2: Calculate the Median

Arrange data in order:

- 15, 17, 18, 19, 20, 21, 22, 25

Find the middle values (8 values, so average the 4th and 5th):

- Middle values: 19 and 20
- Median = $(19 + 20) \div 2 = \mathbf{19.5 \text{ points}}$

Step 3: Calculate the Mode

Count frequency of each value:

- All values appear only once
- **No mode** (or all values are modes)

Step 4: Calculate the Range

- Maximum = 25 points

- Minimum = 15 points
- Range = $25 - 15 = \mathbf{10 \text{ points}}$

Step 5: Calculate the Standard Deviation

Find deviations from mean (19.625):

- $(15 - 19.625) = -4.625$
- $(18 - 19.625) = -1.625$
- $(20 - 19.625) = 0.375$
- $(22 - 19.625) = 2.375$
- $(25 - 19.625) = 5.375$
- $(17 - 19.625) = -2.625$
- $(19 - 19.625) = -0.625$
- $(21 - 19.625) = 1.375$

Square each deviation:

- $(-4.625)^2 = 21.39$
- $(-1.625)^2 = 2.64$
- $(0.375)^2 = 0.14$
- $(2.375)^2 = 5.64$
- $(5.375)^2 = 28.89$
- $(-2.625)^2 = 6.89$
- $(-0.625)^2 = 0.39$
- $(1.375)^2 = 1.89$

Sum of squared deviations:

- $21.39 + 2.64 + 0.14 + 5.64 + 28.89 + 6.89 + 0.39 + 1.89 = 67.87$

Variance:

- $67.87 \div 8 = 8.48$

Standard Deviation:

- $\sqrt{8.48} = \mathbf{2.91 \text{ points}}$ (approximately **2.9 points**)

Summary of Results:

- **Mean = 19.6 points**
- **Median = 19.5 points**
- **Mode = None**
- **Range = 10 points**
- **Standard Deviation = 2.9 points**

Insights about the player's performance:

The player shows solid and steady scoring performance. The mean and median are almost the same, at 19.6 and 19.5, which means there are no significant outliers and the player's performance is balanced. They average nearly 20 points per game, demonstrating that they are a reliable scorer.

The standard deviation of 2.9 points is relatively small compared to the mean, showing good consistency. The player usually scores within 3 points of their average, indicating they are dependable and not prone to extreme hot or cold streaks.

The range of 10 points, from 15 to 25, indicates that the player has a worst game of 15 points and a best game of 25 points. However, most performances cluster around the 19-20 point mark. There is no mode, which means the player doesn't score the exact same amount repeatedly—this is normal for basketball.

Overall, this player is a consistent and reliable scorer. Coaches can count on them to contribute around 20 points per game with minimal variation, making them a valuable asset to the team.